Worksheet: Measures of Spread

Range, Quartiles, and Interquartile Range (IQR)

Year 11 Statistics

2026-08-01

Name: ______________________________ **Date:** ____________

Section 1: Key Concepts & Formula Review

Key Terms

Rawa (Data/Information) Collected values or measurements.

Range The total spread of the data, from the lowest value to the highest value.

Median (Q_2) The exact middle value of an **ordered** data set.

Lower Quartile (Q_1) The middle value of the lower half of the data (the 25th percentile).

Upper Quartile (Q_3) The middle value of the upper half of the data (the 75th percentile).

Interquartile Range (IQR) The spread of the middle 50% of the data. It is a measure of consistency.

Core Formulas

$$\text{Range} = \text{Maximum Value} - \text{Minimum Value}$$

$$\text{Interquartile Range (IQR)} = Q_3 - Q_1$$

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Section 2: Guided Practice

Example Problem

A group of students recorded their paddle times in a Waka Ama sprint session. The times, in seconds, are:

Data Set: `[3, 5, 6, 8, 10, 12, 15]`

1. Find the Minimum and Maximum:

- Minimum = 3
- Maximum = 15

2. Calculate the Range:

- Range = $15 - 3 = 12$

3. Find the Quartiles:

- The data is already ordered.
- **Median** (Q_2): The middle number is **8**.
- **Lower Half**: `[3, 5, 6]` $\longrightarrow$ The middle of this half is **5**. So, $Q_1 = 5$.
- **Upper Half**: `[10, 12, 15]` $\longrightarrow$ The middle of this half is **12**. So, $Q_3 = 12$.

4. Calculate the IQR:

$$\begin{aligned} \text{IQR} &= Q_3 - Q_1 \\ &= 12 - 5 \\ &= 7 \end{aligned}$$

Section 3: Independent Practice

Exercise: Kapa Haka Performance Scores

Two Kapa Haka groups recorded their judges' scores across 7 rounds.

- **Group A:** `[12, 14, 15, 16, 18, 20, 22]`
- **Group B:** `[4, 8, 15, 16, 18, 28, 35]`

Task 1: Analyze Group A

- **Minimum:** ______________________
- **Maximum:** ______________________
- **Range:** Maximum − Minimum = ______________________
- **Median** (Q_2)**:** ______________________
- **Lower Quartile** (Q_1)**:** ______________________
- **Upper Quartile** (Q_3)**:** ______________________
- **IQR:** $Q_3 - Q_1$ = ____________ − ____________ = ____________

Task 2: Analyze Group B

- **Minimum:** ______________________
- **Maximum:** ______________________
- **Range:** Maximum − Minimum = ______________________
- **Median** (Q_2)**:** ______________________

- **Lower Quartile (Q_1):** ______________________
- **Upper Quartile (Q_3):** ______________________
- **IQR:** $Q_3 - Q_1 =$ ____________ $-$ ____________ $=$ ____________

Task 3: Contextual Comparison (Literacy Focus)

Write 2–3 full sentences comparing the spread and consistency of Group A and Group B. Which group performed more consistently? Use values from your calculations (Range or IQR) to justify your answer.

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Section 4: Extension & Challenge

Consider the following score set from a secondary athletics meet. Note that the data set has an even number of values.

Data Set: `[2, 4, 7, 9, 12, 15, 18, 42]`

1. Calculate the Range and IQR for this data set:

- Range = ____________
- IQR = ____________

Hint: For an even-numbered data set, the median (Q_2) is the average of the two middle numbers. The lower half and upper half will then each have 4 numbers. Q_1 and Q_3 will also be averages.

2. Critical Reflection Question:

The score `42` is much higher than the others and could be considered an outlier.

a) How did this outlier affect the **Range** versus the **IQR**?

b) Why is the IQR often a better measure of spread (consistency) than the Range when a data set contains extreme outliers?

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Section 5: Summary Task

Given the ordered data set: `[2, 4, 7, 9, 12, 15, 18]`

Calculate the following:

- **Range:** ____________________
- **Median (Q_2):** ____________________
- **IQR ($Q_3 - Q_1$):** ____________________